Total Sum of Squares: $TSS = \sum\sum x_{ij}^2 - C.F$

$= 76988 - 75757.07 = 1230.93$

Sum of Squares due to Treatments: $SST = \dfrac{\sum_{j=1}^{l} x_{i.}^2}{l} - C.F$

$= \dfrac{227454}{3} - 75757.07$

$= 75818 - 75757.07$

$= 60.93$

Sum of Squares due to Blocks: $SSB = \dfrac{\sum_{i=1}^{k} x_{.j}^2}{k} - C.F$

$= \dfrac{383020}{5} - 75757.07$

$= 76604 - 75757.07$

$= 846.93$

Sum of squares due to error: SSE $= TSS - SST - SSB$

$= 1230.93 - 60.93 - 846.93$

$= 323.07$

ANOVA Table (two-way)

Sources of variation	Sum of squares	Degrees of freedom	Mean sum of squares	F-ratio
Between inspectors	60.93	4	15.23	$F_{0I} = \dfrac{15.23}{40.38} = 0.377$
Between restaurants	846.93	2	423.47	$F_{0R} = \dfrac{423.47}{40.38} = 10.49$
Error	323.07	8	40.38	
Total	1230.93	14		

Step 6 : Critical values

(i) $f_{(4, 8),0.05} = 3.838$ (for inspectors)

(ii) $f_{(2, 8),0.05} = 4.459$ (for restaurants)

Step 7 : Decision

i) As $F_{0I} = 0.377 < f_{(4, 8),0.05} = 3.838$, the null hypothesis is not rejected and we conclude that there is no significant difference among the mean cleanliness scores of inspectors.

ii) As $F_{0R} = 10.49 > f_{(2, 8),0.05} = 4.459$, the null hypothesis is rejected and we conclude that there exists significant difference in atleast one pair of restaurants over their mean cleanliness scores.

12th_Statistics_EM_Unit_3.indd 98 1/28/2020 6:33:46 PM